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[5152]-179

S.E. (II Sem.) (Information Technology) EXAMINATION, 2017
FOUNDATION OF COMPUTER NETWORKS
(2012 PATTERN)

Time : Two Hours

Maximum Marks : 50

- N.B. :—** (i) Answer Q. No. 1 or Q. No. 2, Q. No. 3 or Q. No. 4, Q. No. 5 or Q. No. 6, Q. No. 7 or Q. No. 8.
(ii) Neat diagrams must be drawn wherever necessary.
(iii) Figures to the right indicate full marks.
(iv) Assume suitable data, if necessary.

1. (a) Define computer networks. Discuss various types of networks topologies in computer network. [4]
(b) If a binary signal is sent over a 3 kHz channel whose signal-to-noise ratio is 20 dB, what is the maximum achievable data rate ? [4]
(c) What are the various factors used to evaluate the performance of network ? [4]

Or

2. (a) What are the differences between parallel and serial transmission ? List three different techniques in serial transmission. [6]
(b) Define spread spectrum. Explain FHSS and DSSS. [6]
3. (a) Explain Pulse Code Modulation with suitable diagram. [6]
(b) What is circuit switching ? Explain circuit switching in detail with its advantages and disadvantages. [7]

P.T.O.

Or

4. (a) Explain ISO/OSI model in brief. What are the responsibilities of : [7]
- (i) Physical Layer and
- (ii) Network Layer ?
- (b) What is DSL technology ? Distinguish between ADSL and HDSL. [6]

5. (a) Explain in detail Stop and Wait and Selective Repeat ARQ System. [7]
- (b) Explain Error Detection and Correction in Block Coding. [6]

Or

6. (a) What is hamming distance ? Explain with example. Explain simple parity check code. [8]
- (b) Define piggybacking and its usefulness. [5]
7. (a) Define controlled access and list three protocols in this category. Explain CSMA. [6]
- (b) Compare and contrast HDLC with PPP. Which one is byte-oriented, which one is bit-oriented ? [6]

Or

8. (a) Explain FDMA, TDMA and CDMA in detail. [6]
- (b) Explain the frame format for IEEE 802.3. [6]